chapter #01

Formula.

$d_{prop} = \frac{d}{s}$

Traffic Intensity $= \frac{L.a}{R}$

$d_{trans} = \frac{L}{R}$

L.a = arrival rate of bits

R = service rate of bits.

- $La/R \sim 0$: avg queueing delay → small
- $La/R \sim 1$: " " " → large
- $La/R > 1$: " " " → infinite

cannot be serviced.

$d_{total} = d_{proc} + d_{queue} + d_{prop} + d_{trans}$.

circuit switching:

d_{total} = setup time + d_{prop} + d_{trans} + tear down time

Throughput = $\frac{\text{data sent successfully}}{\text{time taken}}$ (size of file)

Throughput = $\min(R_1, R_2, \ldots R_n)$

bit length $= \frac{m}{R.d_{prop}} = \frac{S}{R}$

S = speed propagation

R = transmission rate

unit : meters / bits.

bandwidth-delay product = R. dprop

chap 2:

utilization = $\frac{\text{avg. data rate}}{\text{link capacity}}$

deze = Internet Delay + Access link Delay + LAN Delay

access link data rate = Total data rate x miss ratio

miss ratio = 1 - hit ratio

avg delay = (hit ratio x d_{cache}) + (miss ratio x d_{origin})

bandwidth saved = Total bandwidth (original) - non cached bandwidth (remaining)

bandwidth saved = Total bandwidth (original) - hit ratio.

Non- persistent HTTP

$$RTT_{non\text{-}persistent} = 2N\ RTT$$

Persistent HTTP

$$RTT_{persistent} = 1 + N\ RTT$$